PCA Numerical.

Range between $-\infty$ & ∞

Covariance Matrix

Correlation — variables

0 to 1

X	Y
1	2
2	4
3	5
4	4
5	5

X : Hours studied

Y : Exam Score.

How to compute correlation value?

$$r = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

$$Cov(x,y) = \frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

Positive correlation.

$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x^{(i)} - \bar{x})^2}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x^{(i)}$$

$\bar{x} = 3$

$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=x}^{n} (x^{(i)} - \bar{x})^2} = \sqrt{\frac{1}{5} \times 10} = \sqrt{2} = 1.414$$

	$x^{(i)} - \bar{x}$	
1	-2	4
2	-1	1
3	0	0
4	1	1
5	2	4
		10

$$\bar{y} = \frac{20}{5} = 4$$

2	-2	4
4	0	0
5	1	1
4	0	0
5	1	1
		6

$$\sigma_y = \sqrt{\frac{1}{5} \times 6} = \sqrt{\frac{6}{5}} = \sqrt{1.2}$$

$= 1.09$

$r = 0$

Summation la.

$$\text{cov}(x, y) = \sum (x^{(i)} - \bar{x}) \times (y^{(i)} - \bar{y})$$

then sum

$$(-2 \times -2) + (-1 \times 0) + (2 \times 1)$$

$$= (4) + (2) = 6$$

$$\text{cov}(x, y) = \frac{1}{5} \times 6 = \boxed{1.2}$$

$$r = \frac{1.2}{1.414 \times 1.09} = \frac{1.2}{1.54} = \boxed{0.778.}$$

$\longrightarrow$

Eigen Values & Eigen vectors

$$\det(A - \lambda I) = 0$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det\left(\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\det \begin{bmatrix} 2-\lambda & 1-0 \\ 1-0 & 2-\lambda \end{bmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0$$

$$(2-\lambda)^2 = 1$$

$$2-\lambda = \pm 1$$

$$\cancel{\lambda = 4}$$

$$2-\lambda = -1 \qquad\qquad 2-\lambda = +1$$

$$2+1 = \lambda \qquad\qquad 2-1 = \lambda$$

$$\lambda = 3 \qquad\qquad \lambda = 1$$

Sub $\lambda = 3$ $\qquad$ ~~ACE~~ $(A - \lambda I)v = 0$

$$\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -v_1 + v_2 \\ v_1 - v_2 \end{bmatrix} = 0$$

$$\boxed{v_1 = v_2} \qquad \text{if } v_1 = 1,\ v_2 = 1 \qquad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\lambda = 1$

$$v_1 = -v_2 \qquad \text{if } v_1 = 1,\ v_2 = -1 \qquad \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Eigen vectors are non - zero vector.

PCA

$$\begin{array}{cccc} & x_1 & x_2 & x_3 \\ X = & \multicolumn{3}{c}{\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix}} \end{array} \qquad d = 3$$

Step 1 : Mean Centering.

$\mu = [\ 1.5 \quad 1.5 \quad 1.5\ ]$

$$X_c = \begin{bmatrix} 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ 0.5 & 1.5 & 0.5 \end{bmatrix}$$

subtract from mean of feature

$2 - 1.5$

Step 2 : Compute covariance matrix

$$\boxed{C = \frac{1}{n} X_c^T X_c}$$

$$\begin{array}{ccc} x_1 & x_2 & x_3 \\ \hline cov(x_1, x_1) & cov(x_1, x_2) & " \\ cov(x_2, x_1) & cov(x_2, x_2) & " \\ " \ x_3, x_1 & x_3, x_2 & x_3, x_3 \\ " \ x_4, x_1 & x_4, x_2 & x_4, x_4 \end{array}$$

$$C = \begin{bmatrix} 1.25 & -.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix}_{3 \times 3}$$

as only 3 feature

Step 3 : Eigen vectors of C are the subspaces of PCA

$\lambda_1 \approx 2 \qquad \lambda_2 \approx 0.75 \qquad \lambda_3 \approx 0$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \qquad u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \qquad u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Dimensionality Reduction: 3D - 2D

Take top 2 eigen vectors

$$W_{2D} = \begin{bmatrix} v_1 & v_2 \end{bmatrix}$$

Project the data points

$$Z = X_c W_{2D} = [\,]$$

Each points will been of 2D

3D → 1D

Take top 1 eigen vector

$$W_{1D} = [v_1]$$

Total variance = $2 + 0.75 + 0$

$= 2.75$

$PC1 = \frac{2}{2.75} = 73\%$ $\quad PC2 = \frac{0.75}{2.75} \simeq 27\%$

if 3D - 2D = 73 + 27 ✓ 100% of PC

3D - 1D = 73% of data.

Hence i would reduce to 2D than 1D.